

הנחיות למצבי החייה

AMERICAN HEART ASSOCIATION

International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care

Circulation

Volume 132(18 suppl 2)

November 3, 2015

Circulation

Volume 122(18 suppl 3)

November 2, 2010

נבחנים יקרים,

מצגת זו מרכזת עבורכם את ההנחיות להחייאה וניהול מצבי חירום לבביים, כפי שפורסמו ע"י American Heart Association בשנים 2010, 2015 (בהנחיות 2015 עודכנו רק חלק מהאלגוריתמים).

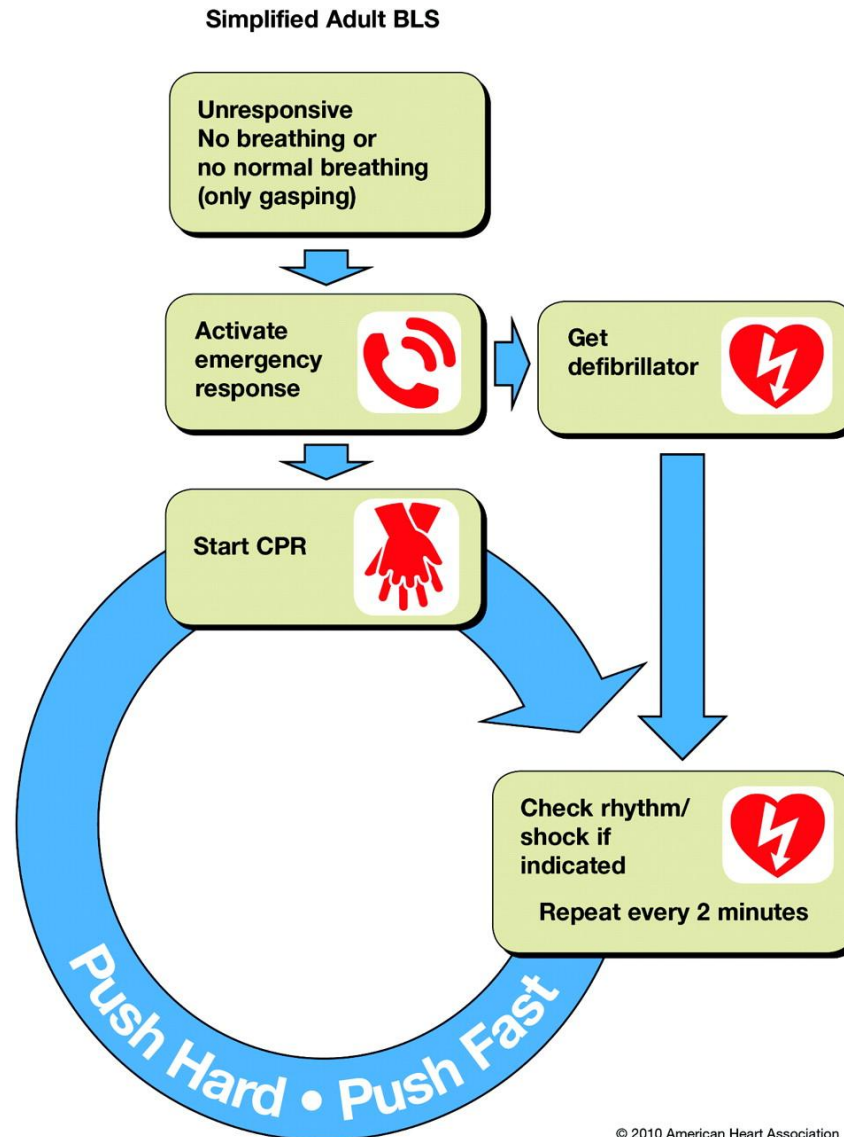
מצבי חירום נוספים (מצוקה נשימתית לסוגיה, בטן חריפה, דהידרציה, כוויות, נשיכות בע"ח ומצבים רבים אחרים) מוצגים במקומות אחרים ברשימת הספרות הרשמית לבחינה, בפרקים הרלבנטיים של ספרי הלימוד הרשמיים לגבי אותו מצב חירום, או בהנחיות קליניות רלבנטיות לנושא.

[הנבחן נדרש לדעת לנהל את כל מצבי החירום שאותם ניתן לפגוש במרפאה!](#)

בברכת בהצלחה,

ועדת הבחינות ברפואת המשפחה

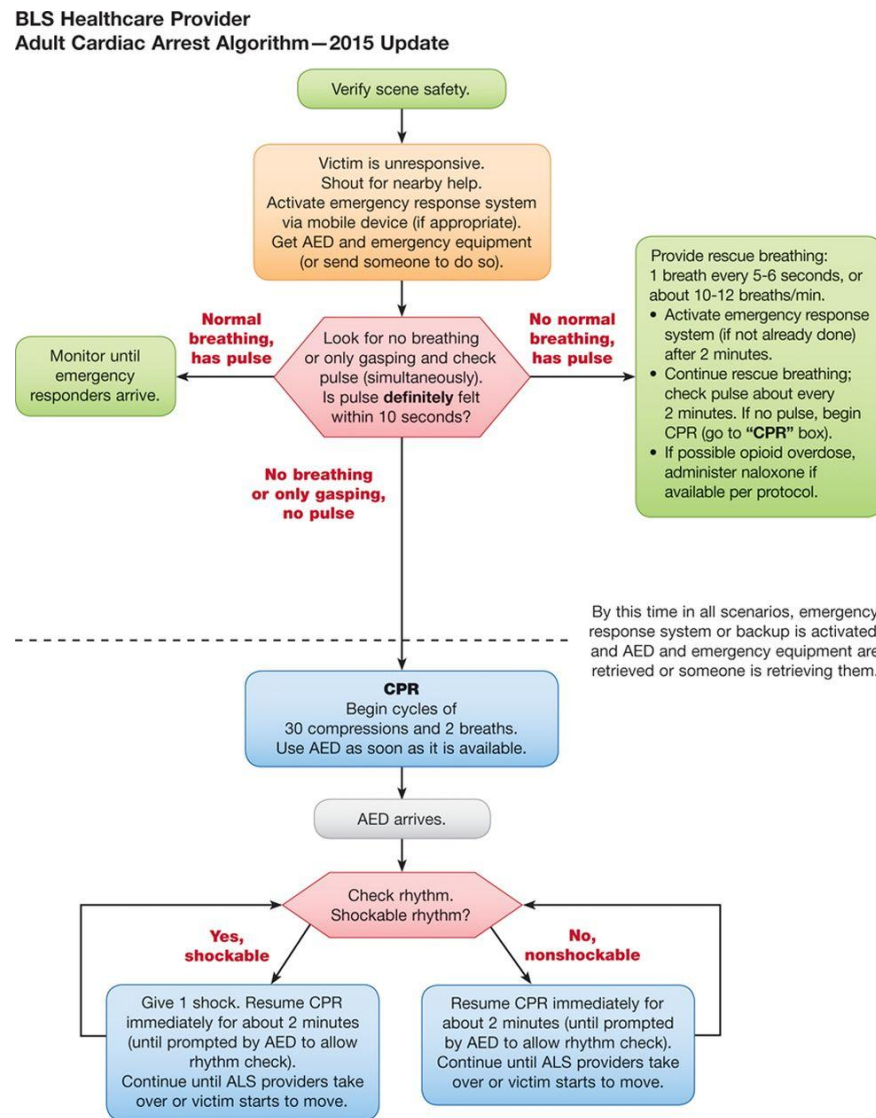
Simplified Adult BLS Algorithm.



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Andrew H. Travers et al. *Circulation*. 2010;122:S676-S684

BLS Healthcare Provider Adult Cardiac Arrest Algorithm—2015 Update.



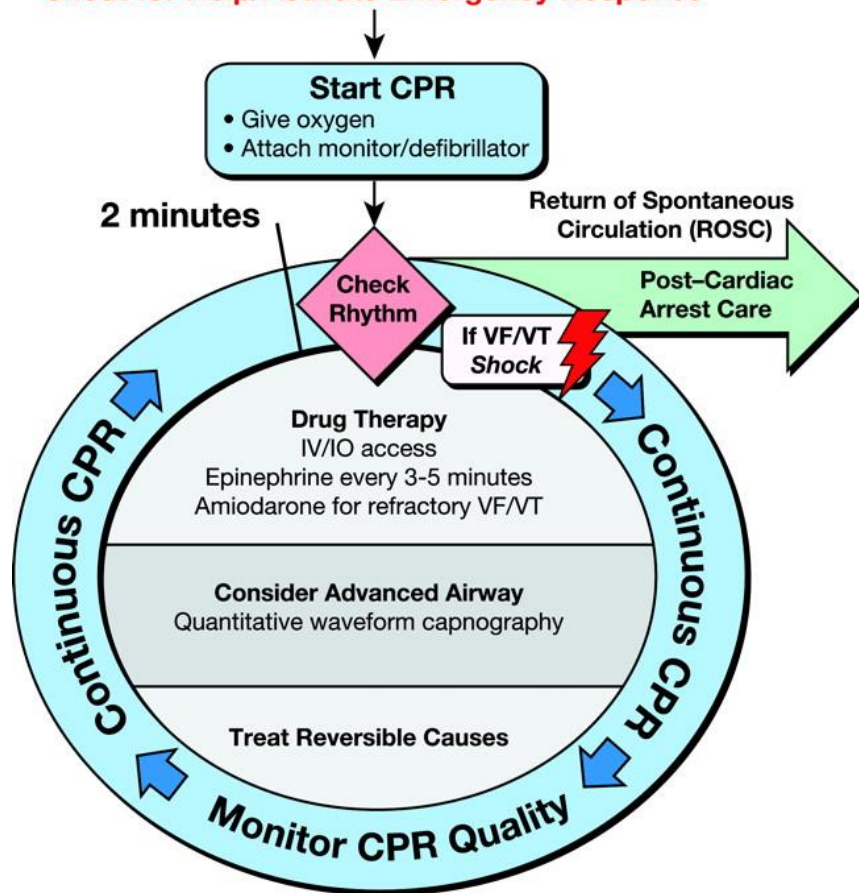
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Monica E. Kleinman et al. *Circulation*. 2015;132:S414-S435

ACLS Cardiac Arrest Circular Algorithm.

Adult Cardiac Arrest

Shout for Help/Activate Emergency Response



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CPR Quality

- Push hard (≥ 2 inches [5 cm]) and fast (≥ 100 /min) and allow complete chest recoil
- Minimize interruptions in compressions
- Avoid excessive ventilation
- Rotate compressor every 2 minutes
- If no advanced airway, 30:2 compression-ventilation ratio
- Quantitative waveform capnography
 - If $PETCO_2 < 10$ mm Hg, attempt to improve CPR quality
- Intra-arterial pressure
 - If relaxation phase (diastolic) pressure < 20 mm Hg, attempt to improve CPR quality

Return of Spontaneous Circulation (ROSC)

- Pulse and blood pressure
- Abrupt sustained increase in $PETCO_2$ (typically ≥ 40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

Shock Energy

- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J

Drug Therapy

- **Epinephrine IV/IO Dose:** 1 mg every 3-5 minutes
- **Vasopressin IV/IO Dose:** 40 units can replace first or second dose of epinephrine
- **Amiodarone IV/IO Dose:** First dose: 300 mg bolus. Second dose: 150 mg.

Advanced Airway

- Supraglottic advanced airway or endotracheal intubation
- Waveform capnography to confirm and monitor ET tube placement
- 8-10 breaths per minute with continuous chest compressions

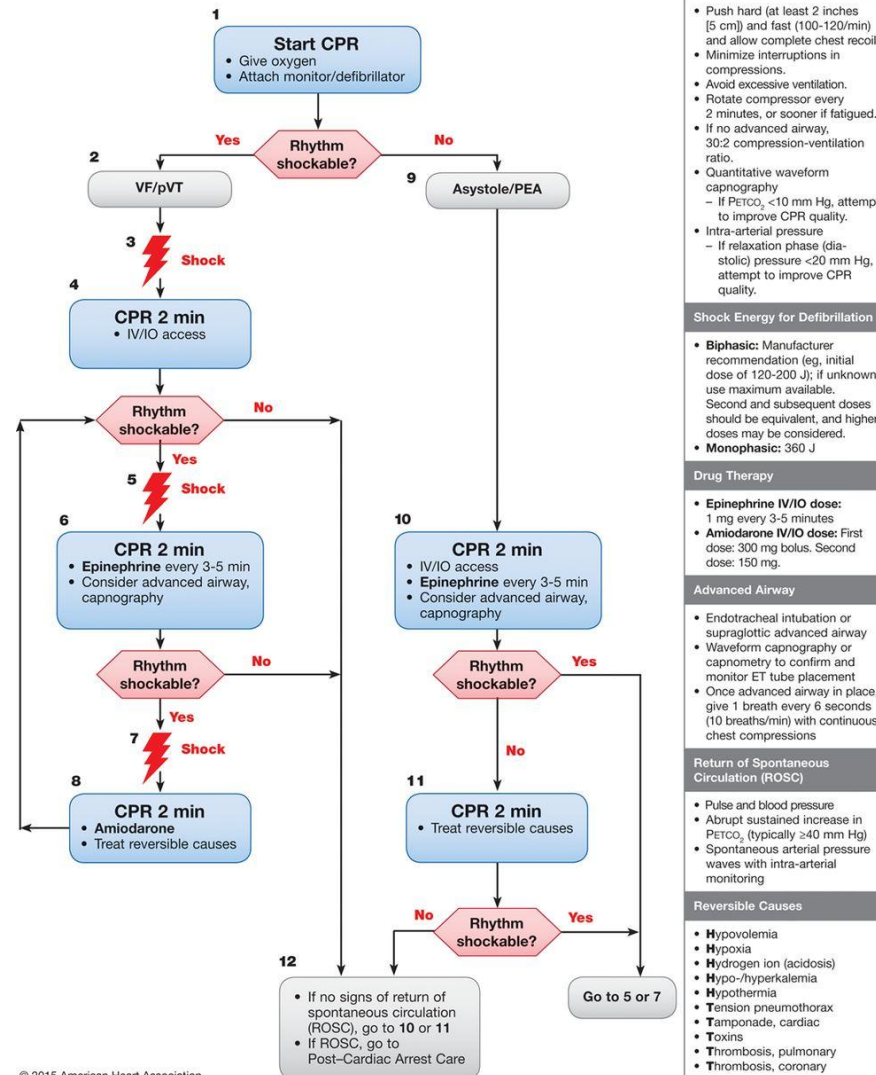
Reversible Causes

- | | |
|---------------------------|-------------------------|
| – Hypovolemia | – Tension pneumothorax |
| – Hypoxia | – Tamponade, cardiac |
| – Hydrogen ion (acidosis) | – Toxins |
| – Hypo-/hyperkalemia | – Thrombosis, pulmonary |
| – Hypothermia | – Thrombosis, coronary |

Robert W. Neumar et al. Circulation. 2010;122:S729-S767

Adult Cardiac Arrest Algorithm—2015 Update.

Adult Cardiac Arrest Algorithm—2015 Update

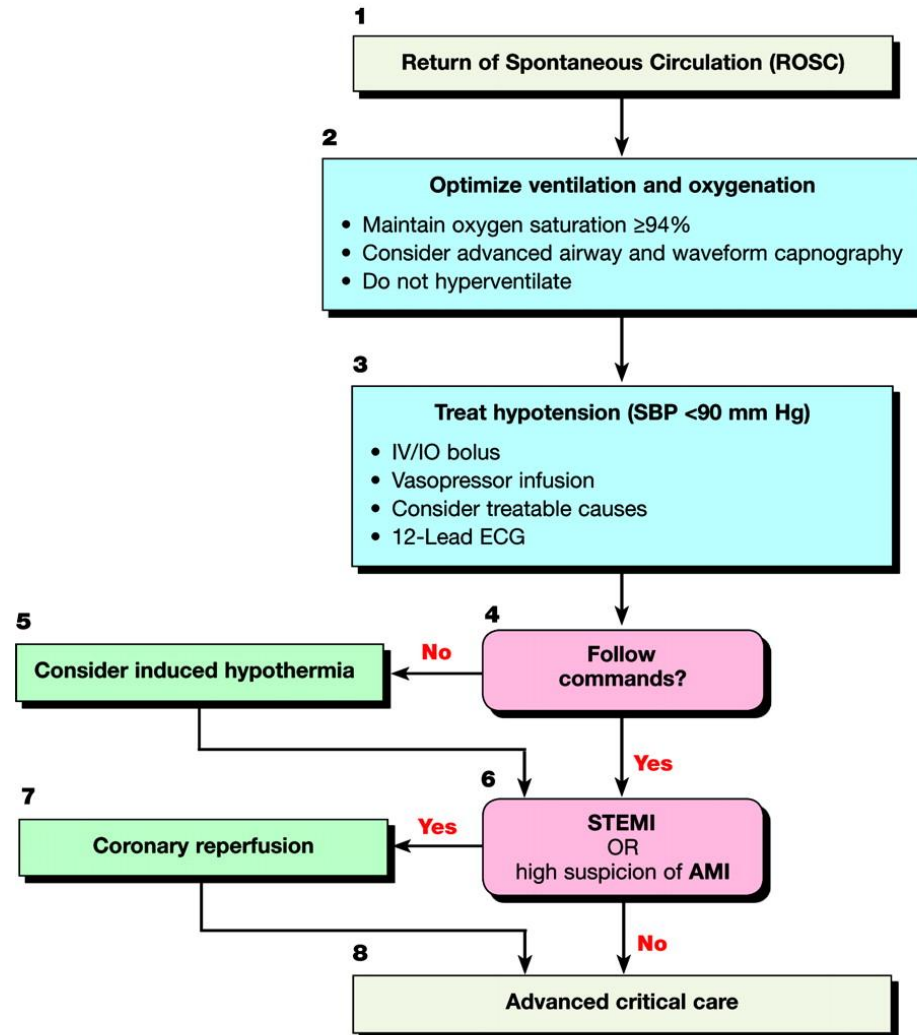


CPR Quality
- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil. - Minimize interruptions in compressions. - Avoid excessive ventilation. - Rotate compressor every 2 minutes, or sooner if fatigued. - If no advanced airway, 30:2 compression-ventilation ratio. - Quantitative waveform capnography - If PETCO₂ <10 mm Hg, attempt to improve CPR quality. - Intra-arterial pressure - If relaxation phase (diastolic) pressure <20 mm Hg, attempt to improve CPR quality.
Shock Energy for Defibrillation
Biphasic: Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered. Monophasic: 360 J
Drug Therapy
Epinephrine IV/IO dose: 1 mg every 3-5 minutes Amiodarone IV/IO dose: First dose: 300 mg bolus. Second dose: 150 mg.
Advanced Airway
- Endotracheal intubation or supraglottic advanced airway - Waveform capnography or capnometry to confirm and monitor ET tube placement - Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions
Return of Spontaneous Circulation (ROSC)
- Pulse and blood pressure - Abrupt sustained increase in PETCO₂ (typically >40 mm Hg) - Spontaneous arterial pressure waves with intra-arterial monitoring
Reversible Causes
- Hypovolemia - Hypoxia - Hydrogen ion (acidosis) - Hypo-/hyperkalemia - Hypothermia - Tension pneumothorax - Tamponade, cardiac - Toxins - Thrombosis, pulmonary - Thrombosis, coronary

Mark S. Link et al. Circulation. 2015;132:S444-S464

Post-cardiac arrest care algorithm.

Adult Immediate Post-Cardiac Arrest Care



Doses/Details

Ventilation/Oxygenation

Avoid excessive ventilation. Start at 10-12 breaths/min and titrate to target PETCO₂ of 35-40 mm Hg. When feasible, titrate FIO₂ to minimum necessary to achieve SpO₂ $\geq 94\%$.

IV Bolus

1-2 L normal saline or lactated Ringer's. If inducing hypothermia, may use 4°C fluid.

Epinephrine IV Infusion:

0.1-0.5 mcg/kg per minute (in 70-kg adult: 7-35 mcg per minute)

Dopamine IV Infusion:

5-10 mcg/kg per minute

Norepinephrine

IV Infusion:

0.1-0.5 mcg/kg per minute (in 70-kg adult: 7-35 mcg per minute)

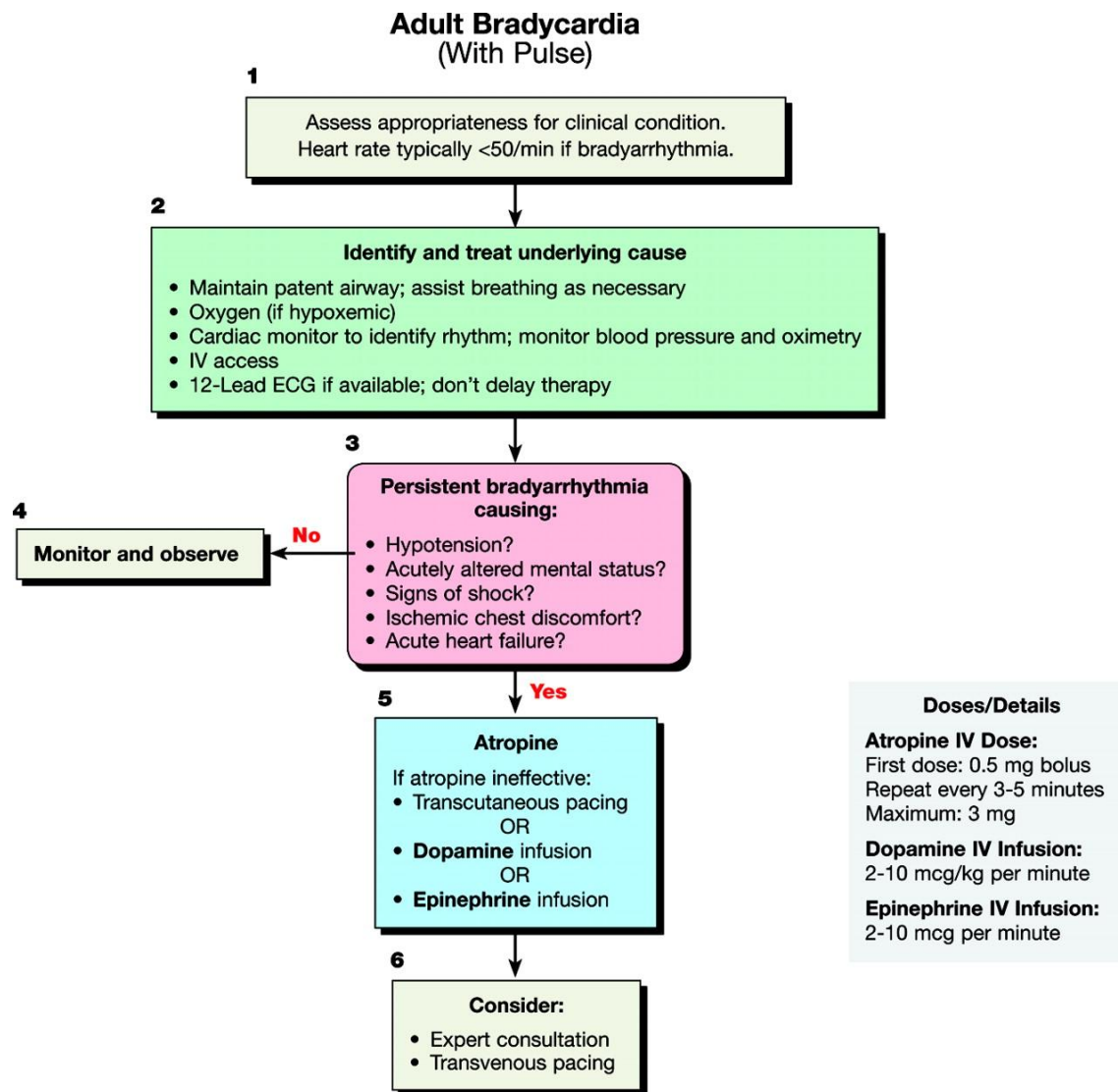
Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

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Mary Ann Peberdy et al. Circulation. 2010;122:S768-S786

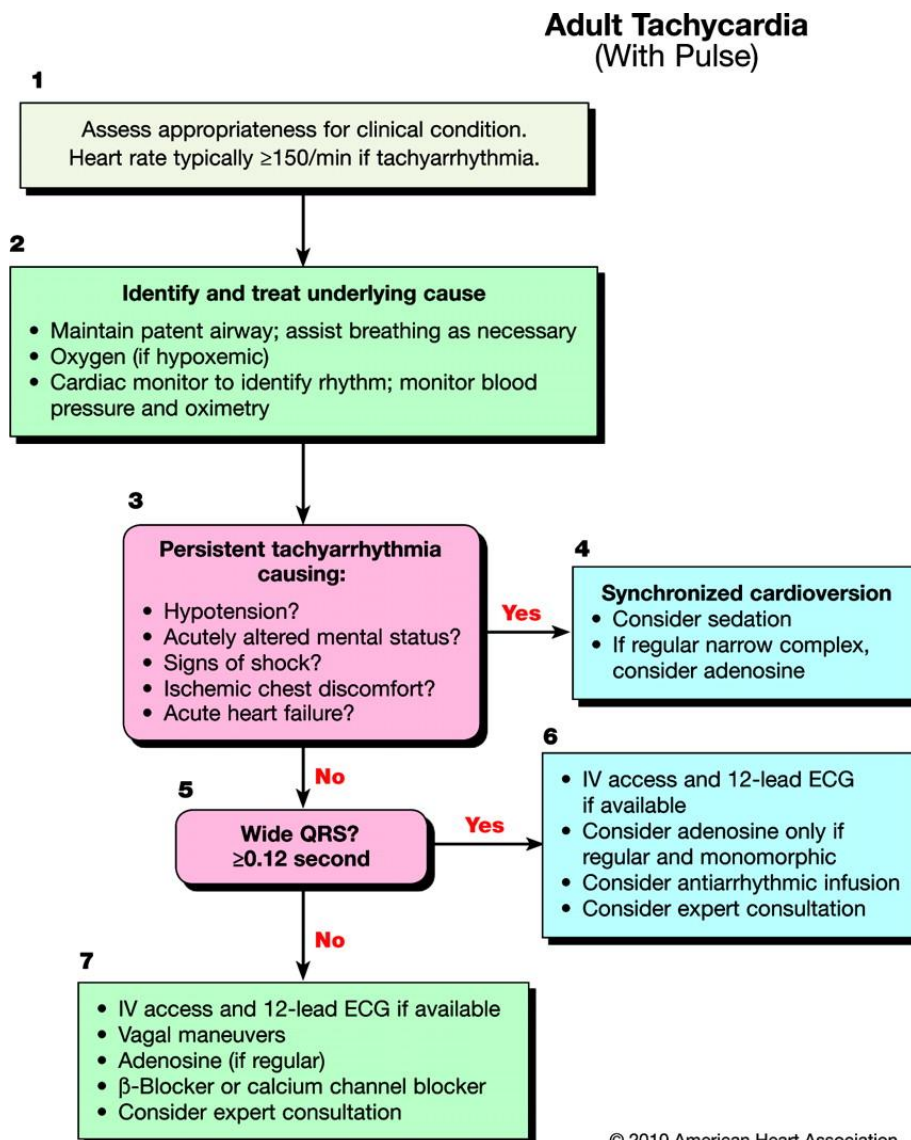
Bradycardia Algorithm.



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Robert W. Neumar et al. Circulation. 2010;122:S729-S767

Tachycardia Algorithm.



Doses/Details

Synchronized Cardioversion

Initial recommended doses:

- Narrow regular: 50-100 J
- Narrow irregular: 120-200 J biphasic or 200 J monophasic
- Wide regular: 100 J
- Wide irregular: defibrillation dose (NOT synchronized)

Adenosine IV Dose:

First dose: 6 mg rapid IV push; follow with NS flush.

Second dose: 12 mg if required.

Antiarrhythmic Infusions for Stable Wide-QRS Tachycardia

Procainamide IV Dose:

20-50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases $>50\%$, or maximum dose 17 mg/kg given. Maintenance infusion: 1-4 mg/min. Avoid if prolonged QT or CHF.

Amiodarone IV Dose:

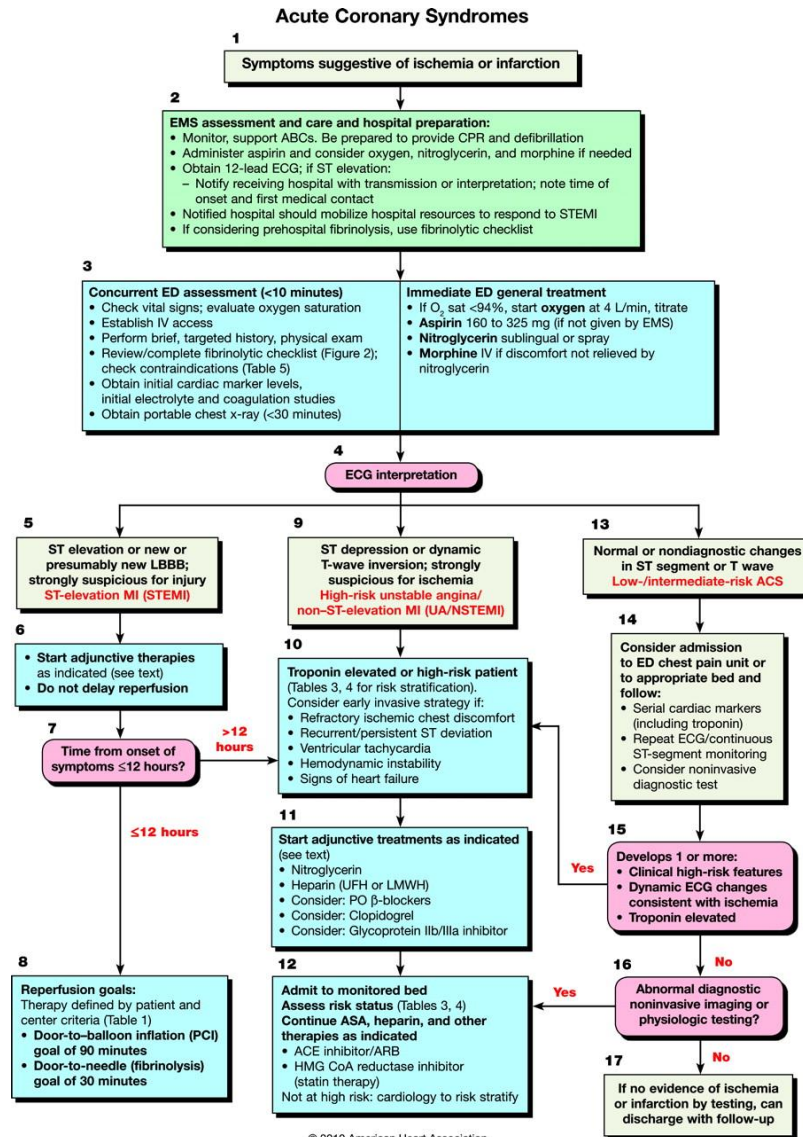
First dose: 150 mg over 10 minutes. Repeat as needed if VT recurs. Follow by maintenance infusion of 1 mg/min for first 6 hours.

Sotalol IV Dose:

100 mg (1.5 mg/kg) over 5 minutes. Avoid if prolonged QT.

Robert W. Neumar et al. Circulation. 2010;122:S729-S767

Acute Coronary Syndromes Algorithm.



Robert E. O'Connor et al. Circulation. 2010;122:S787-S817

Prehospital fibrinolytic checklist.

Prehospital Fibrinolytic Checklist*

Step 1

Has patient experienced chest discomfort for greater than 15 minutes and less than 12 hours?

YES **NO**

Does ECG show STEMI or new or presumably new LBBB?

YES **NO**

STOP

Step 2

Are there contraindications to fibrinolysis?
If **ANY** one of the following is checked **YES**, fibrinolysis MAY be contraindicated.

Systolic BP >180 to 200 mm Hg or diastolic BP >100 to 110 mm Hg	☐ YES	☐ NO
Right vs left arm systolic BP difference >15 mm Hg	☐ YES	☐ NO
History of structural central nervous system disease	☐ YES	☐ NO
Significant closed head/facial trauma within the previous 3 months	☐ YES	☐ NO
Stroke >3 hours or <3 months	☐ YES	☐ NO
Recent (within 2-4 weeks) major trauma, surgery (including laser eye surgery), GI/GU bleed	☐ YES	☐ NO
Any history of intracranial hemorrhage	☐ YES	☐ NO
Bleeding, clotting problem, or blood thinners	☐ YES	☐ NO
Pregnant female	☐ YES	☐ NO
Serious systemic disease (eg, advanced cancer, severe liver or kidney disease)	☐ YES	☐ NO

Step 3

Is patient at high risk?
If **ANY** one of the following is checked **YES**, consider transfer to PCI facility.

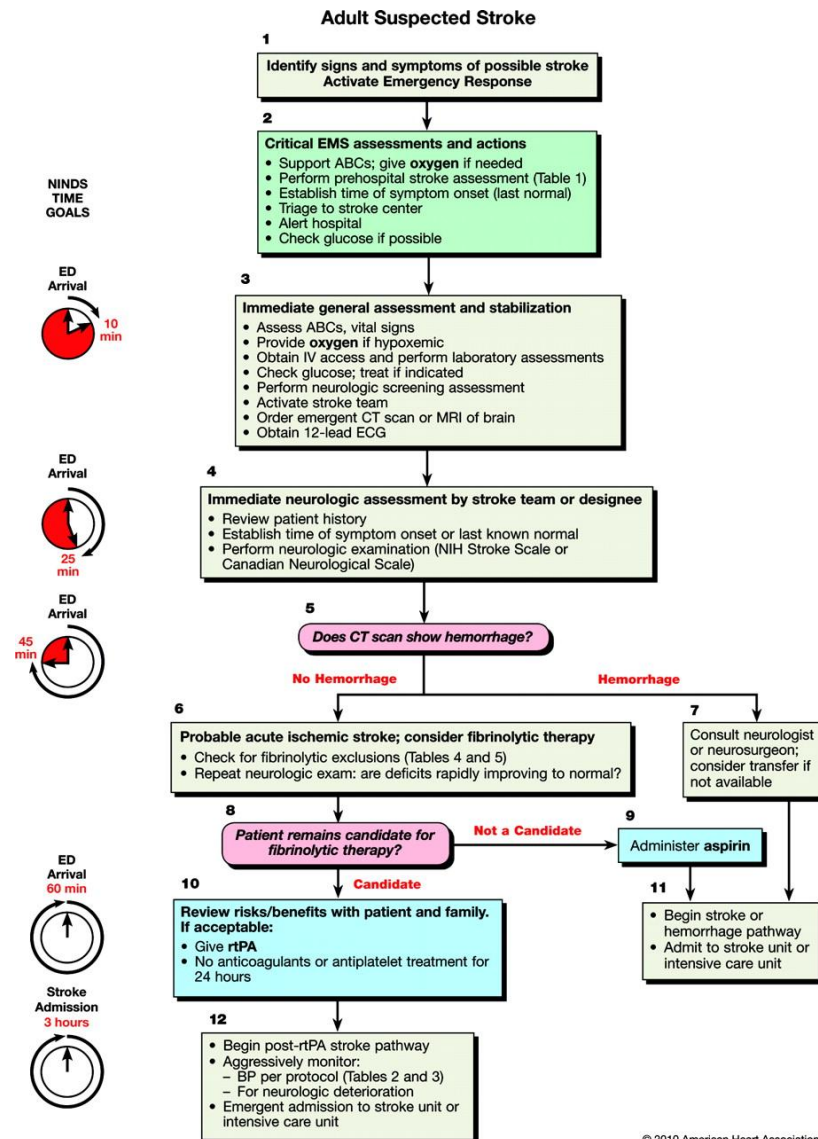
Heart rate $\geq$ 100/min AND systolic BP <100 mm Hg	☐ YES	☐ NO
Pulmonary edema (rales)	☐ YES	☐ NO
Signs of shock (cool, clammy)	☐ YES	☐ NO
Contraindications to fibrinolytic therapy	☐ YES†	☐ NO
Required CPR	☐ YES	☐ NO

*Contraindications for fibrinolytic use in STEMI are viewed as advisory for clinical decision making and may not be all-inclusive or definitive. These contraindications are consistent with the 2004 ACC/AHA Guidelines for the Management of Patients With ST-Elevation Myocardial Infarction.

†Consider transport to primary PCI facility as destination hospital.

Robert E. O'Connor et al. Circulation. 2010;122:S787-S817

Goals for management of patients with suspected stroke.

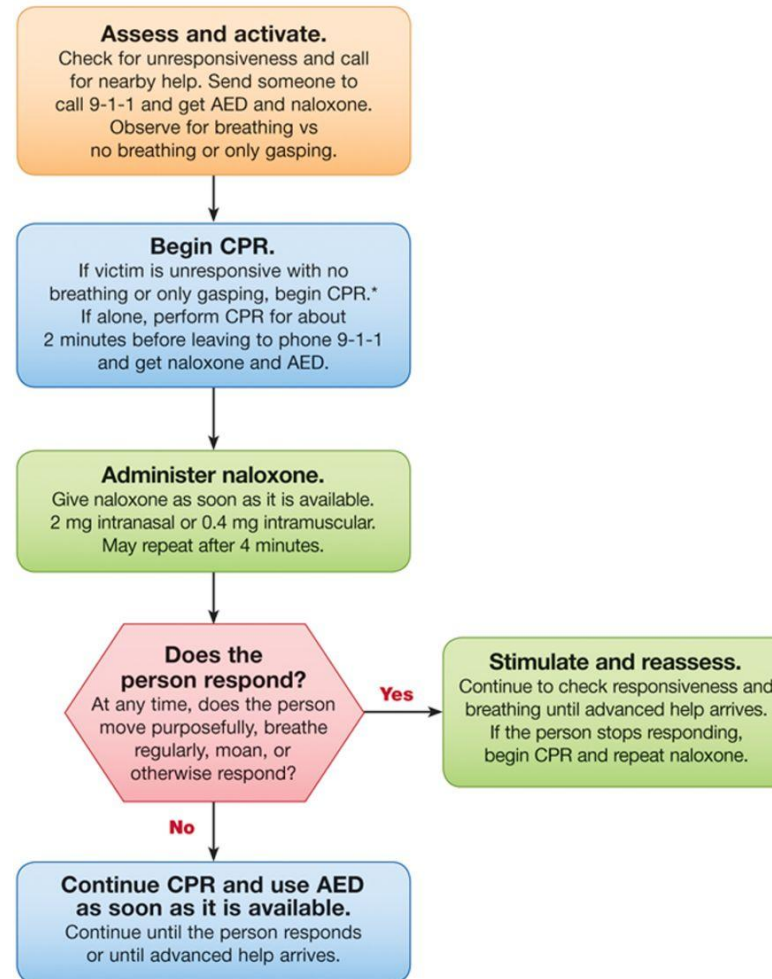


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Edward C. Jauch et al. Circulation. 2010;122:S818-S828

Opioid-Associated Life-Threatening Emergency (Adult) Algorithm.

Opioid-Associated Life-Threatening Emergency (Adult) Algorithm—New 2015



*CPR technique based on rescuer's level of training.

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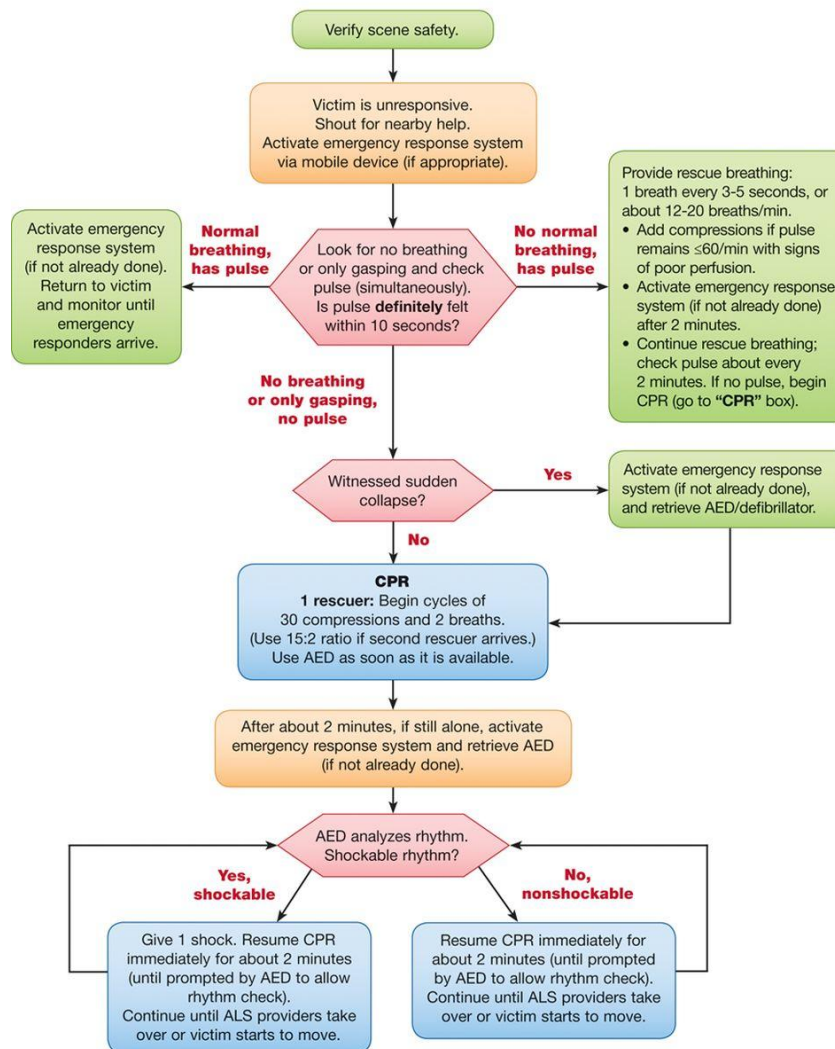
Eric J. Lavonas et al. *Circulation*. 2015;132:S501-S518

Anaphylaxis

ראו הנחיות בקישור הבא מתוך ACLS 2005 (לא עודכן מאז)
http://circ.ahajournals.org/content/112/24_suppl/IV-143

BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for the Single Rescuer—2015 Update.

BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for the Single Rescuer—2015 Update

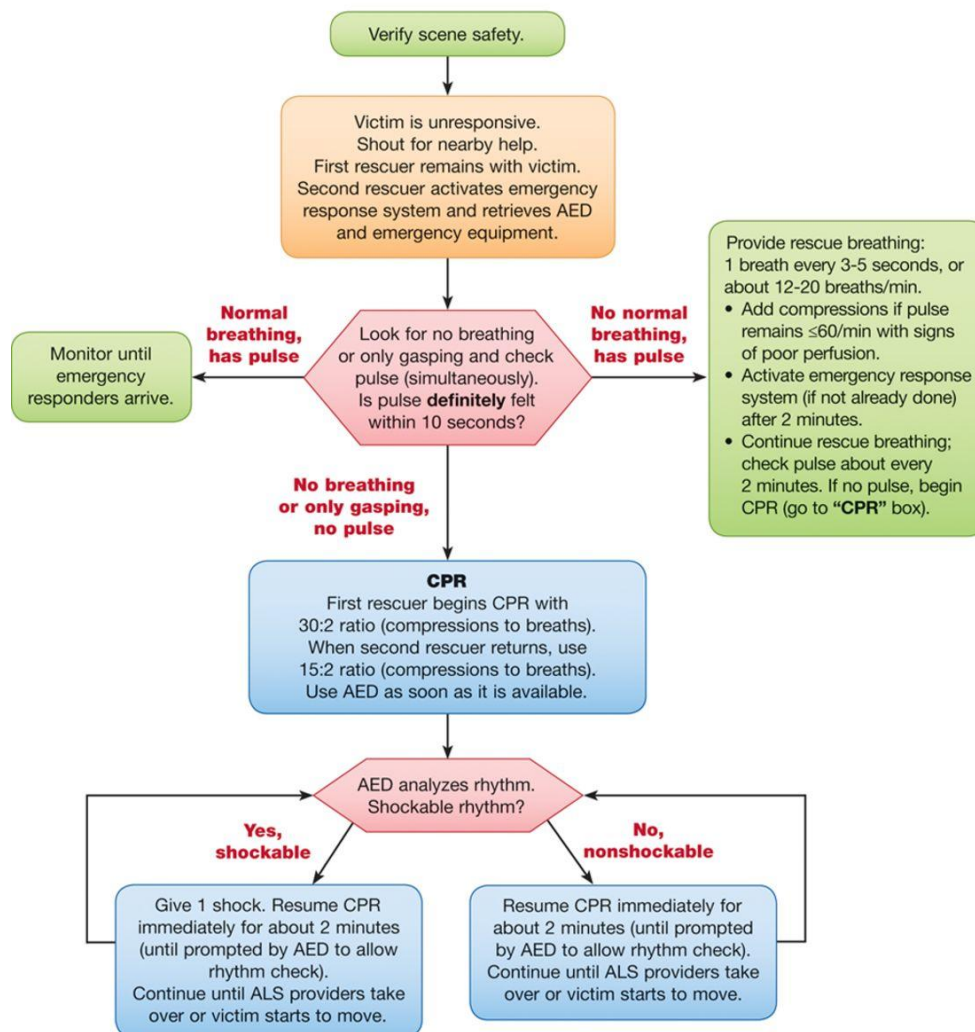


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Dianne L. Atkins et al. Circulation. 2015;132:S519-S525

BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for 2 or More Rescuers—2015 Update.

BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for 2 or More Rescuers—2015 Update

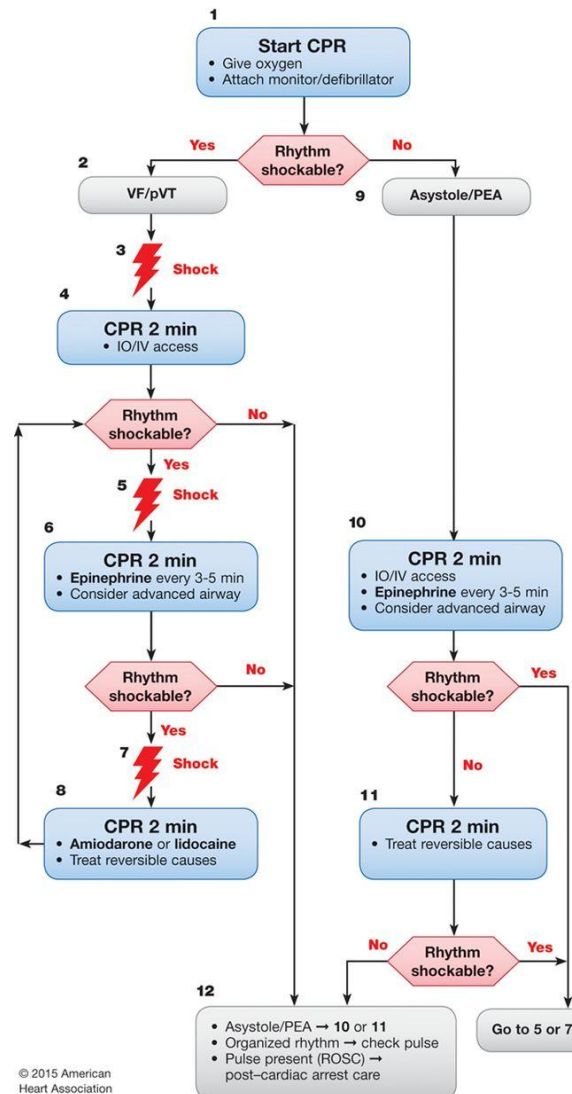


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Dianne L. Atkins et al. *Circulation*. 2015;132:S519-S525

Pediatric Cardiac Arrest Algorithm—2015 Update.

Pediatric Cardiac Arrest Algorithm—2015 Update



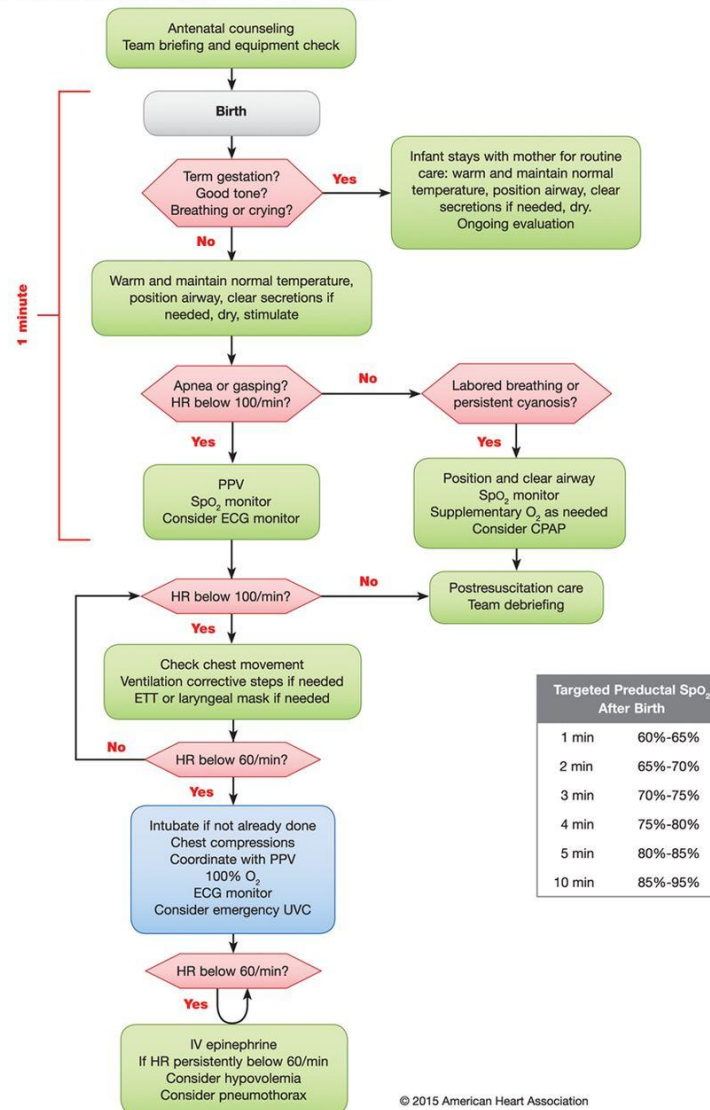
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CPR Quality
• Push hard ($\geq \frac{1}{2}$ of anteroposterior diameter of chest) and fast (100-120/min) and allow complete chest recoil. • Minimize interruptions in compressions. • Avoid excessive ventilation. • Rotate compressor every 2 minutes, or sooner if fatigued. • If no advanced airway, 15:2 compression-ventilation ratio.
Shock Energy for Defibrillation
First shock 2 J/kg, second shock 4 J/kg, subsequent shocks ≥ 4 J/kg, maximum 10 J/kg or adult dose
Drug Therapy
• Epinephrine IO/IV dose: 0.01 mg/kg (0.1 mL/kg of 1:10 000 concentration). Repeat every 3-5 minutes. If no IO/IV access, may give endotracheal dose: 0.1 mg/kg (0.1 mL/kg of 1:1000 concentration). • Amiodarone IO/IV dose: 5 mg/kg bolus during cardiac arrest. May repeat up to 2 times for refractory VF/pulseless VT. • Lidocaine IO/IV dose: Initial: 1 mg/kg loading dose. Maintenance: 20-50 mcg/kg per minute infusion (repeat bolus dose if infusion initiated >15 minutes after initial bolus therapy).
Advanced Airway
• Endotracheal intubation or supraglottic advanced airway • Waveform capnography or capnometry to confirm and monitor ET tube placement • Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions
Return of Spontaneous Circulation (ROSC)
• Pulse and blood pressure • Spontaneous arterial pressure waves with intra-arterial monitoring
Reversible Causes
• Hypovolemia • Hypoxia • Hydrogen ion (acidosis) • Hypoglycemia • Hypo-/hyperkalemia • Hypothermia • Tension pneumothorax • Tamponade, cardiac • Toxins • Thrombosis, pulmonary • Thrombosis, coronary

Allan R. de Caen et al. Circulation. 2015;132:S526-S542

Neonatal Resuscitation Algorithm—2015 Update.

Neonatal Resuscitation Algorithm—2015 Update



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Myra H. Wyckoff et al. Circulation. 2015;132:S543-S560